

TOPOLOGY

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The purpose of this document is to compile common notions in basic topology.

1. BASIC TOPOLOGY

1.1. Topological spaces.

Definition 1.1.1 (Topology). Let X be a set. A **topology on X** is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. For any collection $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ (with I arbitrary), the union $\bigcup_{i \in I} U_i \in \mathcal{T}$,
3. For any finite collection $\{U_1, \dots, U_n\} \subseteq \mathcal{T}$, the intersection $U_1 \cap \dots \cap U_n \in \mathcal{T}$.

If $\mathcal{T}$ is a topology on X , the pair $(X, \mathcal{T})$ is called a **topological space**. Members of $\mathcal{T}$ are called **open sets**.

A subset $C \subseteq X$ is **closed** if its complement $X \setminus C$ is an open set in $\mathcal{T}$.

One very often refers to X as a topological space, omitting the notation of the topology $\mathcal{T}$.

The collection of all topologies on a set X may be denoted by notations such as **Top(X)**, **Top(X)**, or **Top(X)**.

Definition 1.1.2. Let X be a topological space. A subset $U \subseteq X$ is a **clopen subset** if U is both an open set and a closed set in the topology of X .

1.1.1. Basis of a topological space.

Definition 1.1.3. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1) and let $x \in X$.

- An **open neighborhood of x** is any open set $U \in \mathcal{T}$ such that $x \in U$.
- A **neighborhood of x** is a set $N \subseteq X$ for which there exists an open neighborhood $U \in \mathcal{T}$ of x such that $U \subseteq N$.
- A **neighborhood basis** (or **local base**) at x is a nonempty collection $\mathcal{B}_x$ of neighborhoods of x such that for every neighborhood N of x , there exists $B \in \mathcal{B}_x$ with $B \subseteq N$.

The elements of $\mathcal{B}_x$ are said to **form a base of neighborhoods** at x .

Definition 1.1.4. Let X be a set and let $\mathcal{B}$ be a collection of subsets of X . The collection $\mathcal{B}$ is called a **basis** (or **base**) for a topology (Definition 1.1.1) on X if the following two conditions hold:

1. For every $x \in X$, there exists at least one $B \in \mathcal{B}$ such that $x \in B$.

2. For any $B_1, B_2 \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Given such a collection $\mathcal{B}$, the collection $\mathcal{T}$ of all unions of elements of $\mathcal{B}$ defines a topology on X , and it coincides with $\mathcal{T}_{\mathcal{B}}$, the topology generated by $\mathcal{B}$. In other words,

$$\mathcal{T}_{\mathcal{B}} = \{U \subseteq X : \text{for every } x \in U, \text{ there exists } B \in \mathcal{B} \text{ with } x \in B \subseteq U\}.$$

Definition 1.1.5. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1).

- The space $(X, \mathcal{T})$ is said to be *first countable* if every point $x \in X$ has a countable (Definition A.0.1) neighborhood basis (Definition 1.1.3), i.e., there exists a countable collection $\{U_n\}_{n \in \mathbb{N}}$ of open neighborhoods of x such that for any open neighborhood U of x , there exists $n \in \mathbb{N}$ with $U_n \subseteq U$.
- The space $(X, \mathcal{T})$ is said to be *second countable* if there exists a countable basis (Definition 1.1.4) $\mathcal{B} \subseteq \mathcal{T}$ for the topology $\mathcal{T}$, i.e., every open set $U \in \mathcal{T}$ can be written as a union of elements of $\mathcal{B}$.

1.1.2. Separation axioms.

Definition 1.1.6. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1) and let $A \subseteq X$.

- The set A is called a *G_{δ} set* if there exists a countable collection of open sets $\{U_n\}_{n \in \mathbb{N}} \subseteq \mathcal{T}$ such that

$$A = \bigcap_{n \in \mathbb{N}} U_n.$$

- The set A is called an *F_{σ} set* if there exists a countable collection of closed sets $\{F_n\}_{n \in \mathbb{N}}$ such that

$$A = \bigcup_{n \in \mathbb{N}} F_n.$$

Definition 1.1.7 (Separation axioms). Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1).

- $(X, \mathcal{T})$ is *T_0 (Kolmogorov)* if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that, without loss of generality, $x \in U$ and $y \notin U$.
- $(X, \mathcal{T})$ is *T_1 (Fréchet)* if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that $x \in U$, $y \notin U$, and $y \in V$, $x \notin V$.
- $(X, \mathcal{T})$ is *T_2 or Hausdorff* if for every pair of distinct points $x, y \in X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $y \in V$.
- $(X, \mathcal{T})$ is *regular* if it is T_1 and for each point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $F \subseteq V$.
- $(X, \mathcal{T})$ is *T_3 (regular Hausdorff)* if it is T_1 and regular.
- $(X, \mathcal{T})$ is *completely regular* if for each closed set $F \subseteq X$ and $x \notin F$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
- $(X, \mathcal{T})$ is *$T_{3\frac{1}{2}}$ (completely regular Hausdorff)* if it is T_1 and completely regular.
- $(X, \mathcal{T})$ is *normal* if it is T_1 and for each pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $A \subseteq U$ and $B \subseteq V$.
- $(X, \mathcal{T})$ is *T_4 (normal Hausdorff)* if it is T_1 and normal.

- $(X, \mathcal{T})$ is T_5 (completely normal Hausdorff) if it is T_1 and completely normal.
- $(X, \mathcal{T})$ is *perfectly normal* if every closed set is a G_δ (Definition 1.1.6) (countable (Definition A.0.1) intersection of open sets) and the space is normal.
- $(X, \mathcal{T})$ is T_6 (perfectly normal Hausdorff) if it is T_1 and perfectly normal.

1.1.3. Subspaces of a topological space.

Definition 1.1.8 (Subspace topology and subspace). Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$ a subset. The topology induced on Y by $\mathcal{T}$, called the *subspace topology*, is defined as

$$\mathcal{T}_Y := \{U \cap Y : U \in \mathcal{T}\}.$$

The pair $(Y, \mathcal{T}_Y)$ is called a *subspace of $(X, \mathcal{T})$* .

Convention 1.1.9. Given a topological space $(X, \mathcal{T})$ and a subset S of X , one almost always considers S as a subspace of X , i.e. considers S as a topological space equipped with the subspace topology.

Definition 1.1.10 (Discrete topological space). Let X be a set. The *discrete topology on X* is the topology $\mathcal{T}_{\text{disc}}$ defined by

$$\mathcal{T}_{\text{disc}} := \{U : U \subseteq X\},$$

i.e., every subset of X is open. The pair $(X, \mathcal{T}_{\text{disc}})$ is called a *discrete topological space*.

Definition 1.1.11 (Indiscrete (or trivial) topological space). Let X be a set. The *indiscrete topology* (also called the *trivial topology*) on X is the topology $\mathcal{T}_{\text{indisc}}$ defined by

$$\mathcal{T}_{\text{indisc}} := \{\emptyset, X\},$$

i.e., only the empty set and the entire set are open. The pair $(X, \mathcal{T}_{\text{indisc}})$ is called an *indiscrete topological space*.

Definition 1.1.12 (Closure of a subset). Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$ a subset. The *closure of A in X* , denoted by $\overline{A}$, is defined as the intersection of all closed sets containing A , i.e.,

$$\overline{A} := \bigcap \{C \subseteq X : C \text{ is closed and } A \subseteq C\}.$$

Equivalently, $\overline{A}$ is the smallest closed set containing A .

1.2. Properties of topological spaces.

Definition 1.2.1 (Dense subset). Let $(X, \mathcal{T})$ be a topological space. A subset $D \subseteq X$ is called *dense in X* if the closure of D equals X , i.e.,

$$\overline{D} = X.$$

Equivalently, D is dense in X if every non-empty open set $U \in \mathcal{T}$ intersects D , that is,

$$\forall U \in \mathcal{T}, U \neq \emptyset \implies U \cap D \neq \emptyset.$$

Definition 1.2.2. A topological space X is *disconnected* if there exist two disjoint, non-empty open sets (Definition 1.1.1) U and V such that $X = U \cup V$. The space X is *connected* if it is not disconnected.

Definition 1.2.3. Let X be a topological space and x a point in X . The *connected component of x* is the union of all connected (Definition 1.2.2) subspaces (Definition 1.1.8) of X that contain x .

Definition 1.2.4. Let X be a topological space. X is *totally disconnected* if the connected component (Definition 1.2.3) of every point $x \in X$ consists only of the singleton set $\{x\}$.

Proposition 1.2.5. Let X be a compact (Definition 1.2.6) Hausdorff space (Definition 1.1.7). X is totally disconnected (Definition 1.2.4) if and only if the set of clopen subsets (Definition 1.1.2) of X forms a basis for the topology (Definition 1.1.4) of X .

1.2.1. Compact topological spaces.

Definition 1.2.6 (Compact topological space). A topological space $(X, \mathcal{T})$ is *compact* if every open cover of X admits a finite subcover; that is, for every collection $\{U_i\}_{i \in I}$ of open sets in $\mathcal{T}$ such that $X = \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ such that $X = \bigcup_{j=1}^n U_{i_j}$.

Some mathematicians, e.g. algebraic geometers, would refer to this property as *quasi-compactness*.

Here is an alternate, essentially equivalent definition, of compactness:

Definition 1.2.7 (Compactness). Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1). A subset $K \subseteq X$ is *compact* if for every collection $\{U_i\}_{i \in I}$ of open sets such that $K \subseteq \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ with $K \subseteq \bigcup_{j=1}^n U_{i_j}$.

Proposition 1.2.8. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1). A subspace (Definition 1.1.8) $K \subseteq X$ is compact in the sense of Definition 1.2.7 if and only if the space K is compact in the sense of Definition 1.2.6.

Definition 1.2.9 (Locally compact). Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1). X is *locally compact* if for every $x \in X$, there exists an open set $U \in \mathcal{T}$ containing x and a compact set (Definition 1.2.6) $K \subseteq X$ such that $x \in U \subseteq K$.

Definition 1.2.10. Let $(X, \mathcal{T})$ be a topological space and let $A \subseteq X$.

- The *interior of A* , denoted by $\text{int}(A)$, is the union of all open sets contained in A :

$$\text{int}(A) = \bigcup \{U \in \mathcal{T} : U \subseteq A\}.$$

- The *boundary of A* , denoted by ∂A , is defined as

$$\partial A = \overline{A} \setminus \text{int}(A).$$

(Definition 1.1.12) Equivalently, a point $x \in X$ belongs to ∂A if every open neighborhood of x intersects both A and its complement $X \setminus A$.

1.3. Continuous maps of topological spaces.

Definition 1.3.1. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition 1.1.1). A map $f : X \rightarrow Y$ is called **continuous** if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V)$ is an open set in X , that is,

$$\forall V \in \mathcal{T}_Y, f^{-1}(V) \in \mathcal{T}_X.$$

Equivalently, f is continuous if and only if for every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X .

A **map of topological spaces** usually refers to a continuous map between the topological spaces.

The set of continuous maps from X to Y is sometimes denoted by $C(X, Y)$. Other standard notation include $\text{Hom}_{\text{Top}}(X, Y)$ or $\text{Top}(X, Y)$ coming from more general notation for morphisms between objects in a category (Definition B.0.1).

Definition 1.3.2. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition 1.1.1). A function $f : X \rightarrow Y$ is called a **homeomorphism** if it satisfies all of the following:

1. f is bijective (Definition A.0.2);
2. f is continuous (Definition 1.3.1) with respect to $\mathcal{T}_X$ and $\mathcal{T}_Y$; and
3. the inverse map $f^{-1} : Y \rightarrow X$ (Definition A.0.2) is also continuous.

If such a function exists, the spaces X and Y are said to be **homeomorphic**.

Definition 1.3.3 (Pointed topological space). Let X be a topological space (Definition 1.1.1) and let $x_0 \in X$ be a chosen element of X . A **pointed/based (topological) space** is a pair (X, x_0) consisting of the space X together with the distinguished point x_0 , called the **base point of X** . If the base point of a pointed space (X, x_0) is understood, then it may be suppressed from notation; in particular, X may be written as a pointed space as opposed to the full notation of (X, x_0) .

A **morphism of pointed spaces** (or **based map**) or **continuous map** between pointed spaces (X, x_0) and (Y, y_0) is a continuous map (Definition 1.3.1)

$$f : X \rightarrow Y$$

such that $f(x_0) = y_0$.

The collection of pointed spaces with their morphisms form a locally small (Definition B.0.2) category (Definition B.0.1), often called the **category of pointed spaces**. This category is often denoted by notations such as Top_* , $\text{Top}_\bullet$, $\mathbf{Top}_*$, $\mathbf{Top}_\bullet$, etc. The set of continuous maps from pointed spaces X to Y may denoted by notations such as $C_*(X, Y)$, $C_\bullet(X, Y)$, $\text{Top}_*(X, Y)$, $\text{Top}_\bullet(X, Y)$, $\text{Hom}_{\text{Top}_\bullet}(X, Y)$, etc.

Definition 1.3.4 (Preimage of a subset under a map of sets). Let X and Y be sets and let $f : X \rightarrow Y$ be a function. Let $B \subseteq Y$ be a subset of the codomain Y . The **preimage of B under f** (also called the **inverse image of B under f**) is the subset of X defined by

$$f^{-1}(B) := \{x \in X : f(x) \in B\} \subseteq X.$$

If B is a singleton set $\{y\}$, then we often denote $f^{-1}(B)$ by $f^{-1}(y)$.

Definition 1.3.5 (Fiber of a map of topological spaces). Let X and Y be topological spaces (Definition 1.1.1) and let $f : X \rightarrow Y$ be a continuous map (Definition 1.3.1). For a point $y \in Y$, the *fiber of f over y* is the inverse image (Definition 1.3.4) $f^{-1}(y) = f^{-1}(\{y\})$ endowed with the subspace topology (Definition 1.1.8) induced from X . The fiber is also denoted by notations such as $\text{Fib}_f(y)$ or X_y .

1.4. Constructions of topological spaces from other topological spaces.

Definition 1.4.1 (Product of Sets). Let I be a (possibly infinite (Definition A.0.1) but small) index set and let $\{A_i\}_{i \in I}$ be a family of sets indexed by I . The *Cartesian product of the family $\{A_i\}_{i \in I}$* , denoted by $\prod_{i \in I} A_i$, is defined as the set of all tuples/functions

$$\prod_{i \in I} A_i := \{(a_i)_{i \in I} \mid a_i \in A_i \text{ for all } i \in I\},$$

where $(a_i)_{i \in I}$ denotes a function from I to $\bigcup_{i \in I} A_i$ such that $(a_i)_{i \in I}(i) = a_i \in A_i$ for each $i \in I$.

The self product of a set A indexed by I is often denoted by A^I . Note that elements of A^I can be identified with functions $I \rightarrow A$. The finite self product of A taken n times is often denoted by A^n . For finitely many sets $A_1, \dots, A_n$, their Cartesian product is denoted by $A_1 \times \dots \times A_n$. Elements of such a finite product may be written as $(a_1, \dots, a_n)$.

Definition 1.4.2. Let I be an index set and $\{X_i\}_{i \in I}$ be a collection of topological spaces. Let $X = \prod_{i \in I} X_i$ be the set theoretic Cartesian product (Definition 1.4.1) of these spaces.

The *box topology on X* is the topology generated by (Definition 1.1.4) the base (Definition 1.1.4) consisting of all sets of the form $\prod_{i \in I} U_i$, where each U_i is an open subset (Definition 1.1.1) of X_i .

Definition 1.4.3. Let I be an index set and $\{X_i\}_{i \in I}$ be a collection of topological spaces. Let $X = \prod_{i \in I} X_i$ be the set theoretic Cartesian product (Definition 1.4.1) of these spaces.

The *product topology* (or *Tychonoff topology*) on X is the topology generated by (Definition 1.1.4) the base (Definition 1.1.4) consisting of all sets of the form $\prod_{i \in I} U_i$, where each U_i is an open subset (Definition 1.1.1) of X_i and $U_i = X_i$ for all but finitely many indices $i \in I$.

The *product* or the *product space of the X_i* refers to X equipped with the product topology.

For each $j \in I$, let $p_j : \prod_{i \in I} X_i \rightarrow X_j$ denote the canonical projection map. The product topology is the coarsest topology on X such that all projection maps p_j are continuous (Definition 1.3.1).

Proposition 1.4.4. Let I be an index set and $\{X_i\}_{i \in I}$ be a collection of topological spaces. Let $X = \prod_{i \in I} X_i$ be the set theoretic Cartesian product (Definition 1.4.1) of these spaces.

If $X = \prod_{i \in I} X_i$ is endowed with the product topology (Definition 1.4.3), then a function $f : Z \rightarrow X$ from a topological space (Definition 1.1.1) Z is continuous (Definition 1.3.1) if and only if each component function $f_i = p_i \circ f : Z \rightarrow X_i$ is continuous.

This universal property generally fails for the box topology (Definition 1.4.2) when I is infinite (Definition A.0.1).

Theorem 1.4.5 (Tychonoff's theorem). Let I be an index set and let $\{X_i\}_{i \in I}$ be a collection of compact spaces (Definition 1.2.6).

The product space (Definition 1.4.2) $X = \prod_{i \in I} X_i$ is compact in the product topology (Definition 1.1.1).

1.5. Compactly generated.

Definition 1.5.1. (♠ TODO: final topology) A topological space (Definition 1.1.1) X is said to be *compactly generated* (or a *k-space*) if a subset $U \subseteq X$ is open whenever for every compact (Definition 1.2.6) subset $K \subseteq X$, the intersection $U \cap K$ is open in the subspace K . Equivalently, X is compactly generated if and only if the topology of X is the final topology with respect to the collection of inclusions $K \hookrightarrow X$ for compact $K \subseteq X$.

Definition 1.5.2. A topological space (Definition 1.1.1) X is said to be *weakly Hausdorff* if for every continuous map (Definition 1.3.1) $f : K \rightarrow X$ from a compact (Definition 1.2.6) Hausdorff (Definition 1.2.7) space K , the image $\text{Im}(f)$ is closed in X . Equivalently, X is weakly Hausdorff if and only if for every compact Hausdorff K , the induced map $f : K \rightarrow X$ is a closed map.

1.6. Metrics, absolute values, and norms.

1.6.1. Extended metric spaces.

Definition 1.6.1 (Extended Metric). Let M be a set. An *extended metric* on M is a function

$$d : M \times M \rightarrow [0, \infty]$$

such that for all $x, y, z \in M$:

1. **Non-negativity:** $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
2. **Symmetry:** $d(x, y) = d(y, x)$.
3. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$,

again adopting the convention that sums involving ∞ behave so that $a + \infty = \infty$. A set equipped with an extended metric is called an *extended metric space*.

If the range of the extended metric is contained in $[0, \infty)$, then the extended metric is a *metric* in the usual sense and V may be called a *metric space*.

Definition 1.6.2 (Isometry for extended metric spaces). Let (X, d_X) and (Y, d_Y) be extended metric spaces (Definition 1.6.1). A map $f : X \rightarrow Y$ is called an *isometry* if for all $x_1, x_2 \in X$,

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2).$$

An isometry is automatically injective. If, in addition, f is surjective, it is called an *isometric isomorphism*, and X and Y are said to be *isometric*.

Definition 1.6.3 (Extended Metric Induced by an Extended Norm). Let V be a vector space over a field F equipped with an absolute value (Definition 1.7.6)

$$|\cdot| : F \rightarrow [0, \infty),$$

and let $\|\cdot\| : V \rightarrow [0, \infty]$ be an extended norm on V (Definition 1.7.7). Then the *extended metric induced by the extended norm* is the function

$$d : V \times V \rightarrow [0, \infty]$$

defined by

$$d(x, y) := \|x - y\|.$$

It is indeed an extended metric. If $\|\cdot\|$ is a norm, then d is a metric.

Definition 1.6.4 (Topology induced by an extended metric). Let (X, d) be an extended metric space (Definition 1.6.1).

1. Given $\varepsilon \in (0, \infty)$ and $x \in X$, the set

$$B(x, \varepsilon) := \{y \in X : d(x, y) < \varepsilon\}$$

is called the *open ball of radius ε centered at x* .

2. The *topology induced by d on X* is the topology (Definition 1.1.1) defined by declaring a subset $U \subseteq X$ to be open if for every $x \in U$, there exists $\varepsilon \in (0, \infty)$ such that $B(x, \varepsilon)$ is contained in U .

1.7. Convergence in extended metric spaces.

Definition 1.7.1 (Convergence and Limits in an Extended Metric Space). Let (X, d) be an extended metric space (Definition 1.6.1). A sequence $(x_n)_{n=1}^\infty$ in X *converges to a point $x \in X$* if for every $\varepsilon > 0$, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$,

$$d(x_n, x) < \varepsilon.$$

In that case, x is called the *limit of the sequence (x_n)* , and we write

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{or} \quad x_n \rightarrow x$$

Definition 1.7.2 (Cauchy sequence in an extended metric space). Let (M, d) be an extended metric space (Definition 1.6.1). A sequence $(x_n)_{n \in \mathbb{N}}$ in M is called a *Cauchy sequence* if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_m, x_n) < \epsilon$ (and thus, in particular, all pairwise distances $d(x_m, x_n)$ are finite for $m, n \geq N$).

Definition 1.7.3. Let (X, d) be an extended metric space (Definition 1.6.1). The space (X, d) is said to be *complete* if every Cauchy sequence in X (Definition 1.7.2) converges to a point in X (Definition 1.7.1); that is, for every sequence $(x_n)_{n \in \mathbb{N}}$ in X such that for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ with $d(x_m, x_n) < \varepsilon$ for all $m, n \geq N$, there exists $x \in X$ such that

$$\lim_{n \rightarrow \infty} d(x_n, x) = 0.$$

Definition 1.7.4. Let (X, d) be an extended metric space (Definition 1.6.1). A *completion of (X, d)* is a pair $(\hat{X}, \hat{d})$ together with an isometric (Definition 1.6.2) embedding $i : X \rightarrow \hat{X}$, where:

- $(\widehat{X}, \widehat{d})$ is a complete extended metric space (Definition 1.7.1),
- $i(X)$ is dense in $\widehat{X}$ (Definition 1.2.1) (Definition 1.2.1) (with respect to the topologies induced by (Definition 1.6.4) the extended metrics),
- $d(x, y) = \widehat{d}(i(x), i(y))$ for all $x, y \in X$.

(♠ TODO: prove that completions of extended metric spaces exist) As such a space $\widehat{X}$ exist, and is unique up to unique isometry, we may refer to the space $\widehat{X}$ as called *the completion of X (with respect to d)*.

Proposition 1.7.5. Let (X, d) be an extended metric space (Definition 1.6.1). A completion (Definition 1.7.4) $(\widehat{X}, \widehat{d})$ of (X, d) exists and is unique up to unique isometry. Moreover, if (X, d) is a metric space, then so is $(\widehat{X}, \widehat{d})$.

Proof. (♠ TODO: :) actually complete this proof Let $\mathcal{C}$ be the set of all Cauchy sequences in (X, d) :

$$\mathcal{C} = \{(x_n)_{n \in \mathbb{N}} : (x_n) \text{ is a Cauchy sequence in } (X, d)\},$$

and define a relation $\sim$ on $\mathcal{C}$ by

$$(x_n) \sim (y_n) \iff \lim_{n \rightarrow \infty} d(x_n, y_n) = 0.$$

This relation is well-defined and in fact an equivalence relation. (♠ TODO: Show that the relation is well-defined and an equivalence relation) Let $\widehat{X} = \mathcal{C} / \sim$ and let $\widehat{d} : \widehat{X} \times \widehat{X} \rightarrow [0, \infty]$ be defined by

$$\widehat{d}([(x_n)], [(y_n)]) := \lim_{n \rightarrow \infty} d(x_n, y_n).$$

This limit exists as a value in $[0, \infty]$ and is independent of representative sequences chosen.

(♠ TODO: Demonstrate why the limit exists and is well defined)

It can be verified that $\widehat{d}$ is an extended metric on $\widehat{X}$ and that $(\widehat{X}, \widehat{d})$ is complete by construction.

Define an embedding

$$i : X \rightarrow \widehat{X}, \quad i(x) := [(x, x, x, \dots)],$$

which is an isometry:

$$\widehat{d}(i(x), i(y)) = \lim_{n \rightarrow \infty} d(x, y) = d(x, y).$$

Moreover, $i(X)$ is dense in $\widehat{X}$ because any equivalence class $[(x_n)]$ of a Cauchy sequence can be approximated arbitrarily closely by points $i(x_n)$ in the image of X .

Uniqueness:

Suppose $(\widehat{X}', \widehat{d}')$ is another completion of (X, d) with isometric embedding $i' : X \rightarrow \widehat{X}'$.

By the universal property of completions, there exists a unique isometry

$$\Phi : \widehat{X} \rightarrow \widehat{X}'$$

such that $\Phi \circ i = i'$.

The map Φ is onto since $i'(X)$ is dense in $\widehat{X'}$ and Φ preserves limits of Cauchy sequences.

Therefore, Φ is a unique isometry between the completions, and the completion is unique up to unique isometry. $\square$

1.7.1. Absolute values on fields.

Definition 1.7.6 (Absolute Value on a Field). Let F be a field. An **absolute value on F** is a function

$$|\cdot| : F \rightarrow \mathbb{R}_{\geq 0}$$

that satisfies the following properties for all $a, b \in F$:

1. Non-negativity: $|a| \geq 0$,
2. Positive-definiteness: $|a| = 0 \iff a = 0$,
3. Multiplicativity: $|ab| = |a| \cdot |b|$,
4. Triangle inequality: $|a + b| \leq |a| + |b|$.

Here, 0 denotes the additive identity of the field F , and the codomain $\mathbb{R}_{\geq 0}$ consists of non-negative real numbers.

1.7.2. Norms on vector spaces over fields with absolute values.

Definition 1.7.7 (Extended Norm). Let V be a vector space over a field F equipped with an absolute value (Definition 1.7.6)

$$|\cdot| : F \rightarrow [0, \infty).$$

An **extended norm on V** is a function

$$\|\cdot\| : V \rightarrow [0, \infty]$$

satisfying for all $x, y \in V$ and all scalars $\alpha \in F$:

1. **Positive definiteness:** $\|x\| = 0$ if and only if $x = 0$.
2. **Homogeneity:** $\|\alpha x\| = |\alpha| \cdot \|x\|$.
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$,

where arithmetic is extended to allow sums involving ∞ with the convention that $a + \infty = \infty$ for any $a \in [0, \infty]$. A vector space with an extended norm over a field with an absolute value is called an **extended normed (vector) space**.

If the range of the extended norm is contained in $[0, \infty)$, then the extended norm is a **norm** in the usual sense and V may be called a **normed (vector) space**.

Definition 1.7.8 (Topology induced by a norm). Let V be a vector space over a field K with absolute value (Definition 1.7.6) $|\cdot|$, and let $\|\cdot\|$ be an extended norm on V (Definition 1.7.7). The **topology induced by the extended norm $\|\cdot\|$ on V** is defined by declaring a subset $U \subseteq V$ to be open if for every $x \in U$, there exists $\varepsilon > 0$ such that

$$B(x, \varepsilon) := \{y \in V : \|y - x\| < \varepsilon\}$$

is contained in U . The set $B(x, \varepsilon)$ is called the *open ball of radius ε around x* . The collection of all such open sets forms a topology on V .

Equivalently, the topology on V induced by the extended norm $\|\cdot\|$ is the topology on V induced by (Definition 1.6.4) the extended metric $d : V \times V \rightarrow [0, \infty]$ induced by $\|\cdot\|$ (Definition 1.6.3).

1.7.3. Inner product on a vector space over the reals or complex numbers.

Definition 1.7.9. An *inner product on a vector space V over a field $\mathbb{F}$* (either $\mathbb{R}$ or $\mathbb{C}$) is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$$

satisfying, for all $u, v, w \in V$ and all scalars $\alpha \in \mathbb{F}$:

1. **Conjugate symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$
2. **Linearity in the first argument:** $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$
3. **Positive-definiteness:** $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ if and only if $v = 0$

A vector space over $\mathbb{F}$ equipped with such an inner product is called an *inner product space*.

Definition 1.7.10. Let V be a vector space over a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $\langle \cdot, \cdot \rangle$ be an inner product (Definition 1.7.9). The *norm induced by $\langle \cdot, \cdot \rangle$* is the norm (Definition 1.7.7)

$$\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$$

defined by

$$\|v\| = \sqrt{\langle v, v \rangle}$$

for any $v \in V$.

1.8. Euclidean space.

Definition 1.8.1. For a positive integer n , let $\mathbb{R}^n$ denote the n -fold Cartesian product of the real line $\mathbb{R}$ with itself:

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R} \text{ for all } i = 1, \dots, n\}.$$

The set $\mathbb{R}^n$ is called *Euclidean n -space*. A point $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ is associated with the *Euclidean norm*

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}.$$

The corresponding metric $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow [0, \infty)$ is given by

$$d(x, y) = \|x - y\|.$$

This metric induces the standard topology on $\mathbb{R}^n$, called the *Euclidean topology*.

2. HOMOTOPY

The notions of homotopy have the following variants:

1. Absolute, on unpointed spaces
2. Relative, on unpointed spaces
3. Absolute, on pointed spaces
4. Relative, on pointed spaces.

Definition 2.0.1 (Homotopy of maps of topological spaces). Let X and Y be topological spaces and let $K \subseteq X$ be a subset. Let $C(X, Y)$ denote the set of all continuous maps $f : X \rightarrow Y$.

1. A *homotopy between two maps $f, g \in C(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $x \in K$ and $t \in [0, 1]$,

$$H(x, t) = f(x) = g(x).$$

If such an H exists, we say f and g are *homotopic relative to K* , and we write $f \simeq g \text{ rel } K$; this is an equivalence relation.

A *homotopy between two maps $f, g \in C(X, Y)$* is simply a homotopy relative to $\emptyset$. We write we write $f \simeq g$ if a homotopy between them exists.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.3.3) and let $K \subseteq X$ be a subset with $x_0 \in K$. Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ satisfying $f(x_0) = y_0$.

A *homotopy of based maps $f, g \in C_*(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

in particular fixing the basepoint throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

If such an H exists, we say f and g are *based homotopic relative to K* , and we write $f \simeq g \text{ rel } K$. This is an equivalence relation.

A *homotopy of based maps $f, g \in C_*(X, Y)$* without relative condition is the special case $K = \{x_0\}$ and is called a *homotopy of based maps* or *based homotopy*. We write $f \simeq g$ if such a homotopy exists.

Definition 2.0.2 (Homotopy equivalence). Let X and Y be topological spaces and let $K \subseteq X$.

1. A continuous map (Definition 1.3.1) $f : X \rightarrow Y$ is a **homotopy equivalence relative to K** if there exists a continuous map $g : Y \rightarrow X$ such that

$$g \circ f \simeq \text{id}_X \text{ rel } K \quad \text{and} \quad f \circ g \simeq \text{id}_Y \text{ rel } f(K).$$

(Definition 2.0.1) That is, there exist homotopies (Definition 2.0.1) $H : X \times [0, 1] \rightarrow X$ and $G : Y \times [0, 1] \rightarrow Y$ satisfying

$$\begin{aligned} H(x, 0) &= (g \circ f)(x), & H(x, 1) &= x, & H(k, t) &= k \text{ for all } k \in K, t \in [0, 1], \\ G(y, 0) &= (f \circ g)(y), & G(y, 1) &= y, & G(f(k), t) &= f(k) \text{ for all } k \in K, t \in [0, 1]. \end{aligned}$$

In this case, the map g is called a **homotopy inverse relative to K of f** , and the pairs (X, K) and $(Y, f(K))$ are said to be **homotopy equivalent relative to K** , denoted $(X, K) \simeq (Y, f(K)) \text{ rel } K$. This is an equivalence relation.

A continuous map $f : X \rightarrow Y$ is a **homotopy equivalence** if it is a homotopy equivalence relative to $\emptyset$. A **homotopy inverse of f** is then simply a homotopy inverse of f relative to $\emptyset$. In this case, and the spaces X and Y are said to be **homotopy equivalent**, denoted $X \simeq Y$. This is an equivalence relation.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.3.3) and let $K \subseteq X$ be a subset with $x_0 \in K$.

A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces is a **homotopy equivalence relative to K** if there exists a continuous map $g : (Y, y_0) \rightarrow (X, x_0)$ such that

$$g \circ f \simeq \text{id}_X \text{ rel } K \quad \text{and} \quad f \circ g \simeq \text{id}_Y \text{ rel } f(K),$$

where the homotopies (Definition 2.0.1) are homotopies of based maps relative to K and $f(K)$, respectively.

That is, there exist homotopies of based maps $H : X \times [0, 1] \rightarrow X$ and $G : Y \times [0, 1] \rightarrow Y$ satisfying

$$\begin{aligned} H(x, 0) &= (g \circ f)(x), & H(x, 1) &= x, & H(k, t) &= k \text{ for all } k \in K, t \in [0, 1], \\ G(y, 0) &= (f \circ g)(y), & G(y, 1) &= y, & G(f(k), t) &= f(k) \text{ for all } k \in K, t \in [0, 1], \end{aligned}$$

with $H(x_0, t) = x_0$ and $G(y_0, t) = y_0$ for all t .

The map g is called a **homotopy inverse relative to K of f** , and the pairs (X, K) and $(Y, f(K))$ are said to be **homotopy equivalent relative to K** , denoted $(X, K) \simeq (Y, f(K)) \text{ rel } K$. This is an equivalence relation.

A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces is a **homotopy equivalence** if it is a homotopy equivalence relative to $\{x_0\}$. A **homotopy inverse of f** is then simply a homotopy inverse relative to $\{x_0\}$. In this case, the pointed spaces (X, x_0) and (Y, y_0) are said to be **homotopy equivalent**, denoted $(X, x_0) \simeq (Y, y_0)$.

Definition 2.0.3 (Homotopy class of maps relative to a subset). Let X and Y be topological spaces (Definition 1.1.1) and let $K \subseteq X$. Let $C(X, Y)$ denote the set of all continuous maps (Definition 1.3.1) $f : X \rightarrow Y$.

1. Two maps $f, g \in C(X, Y)$ are said to be in the same **homotopy class relative to K** if there exists a homotopy relative to K (Definition 2.0.1)

$$H : X \times [0, 1] \rightarrow Y$$

such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and

$$H(k, t) = f(k) = g(k) \quad \text{for all } k \in K, t \in [0, 1].$$

The *homotopy class of maps relative to K* containing a map $f : X \rightarrow Y$ is denoted by $[f]_K$.

Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\emptyset$.

The *homotopy class of maps* containing a map $f : X \rightarrow Y$ is denoted by $[f]$.

The set of homotopy classes of maps may often be denoted by $[X, Y]$.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.3.3) and let $K \subseteq X$ be a subset containing x_0 . Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ with $f(x_0) = y_0$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class relative to K* if there exists a homotopy of based maps relative to K

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

particularly ensuring the basepoint is fixed throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

The *homotopy class relative to K* containing $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]_K$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\{x_0\}$.

The *homotopy class* containing a map $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]$.

The set of homotopy classes of pointed maps $(X, x_0) \rightarrow (Y, y_0)$ may often be denoted by $[(X, x_0), (Y, y_0)]$ or by $[X, Y]$ if the base points are clear.

Definition 2.0.4. The *homotopy category of topological spaces*, denoted $\mathbf{hTop}$, is the category whose objects are topological spaces (Definition 1.1.1) and whose morphisms are homotopy classes (Definition 2.0.3) of continuous maps (Definition 1.3.1). In other words, for objects X and Y , the set of morphisms is defined as $\text{Hom}_{\mathbf{hTop}}(X, Y) = [X, Y] = C(X, Y) / \simeq$.

Proposition 2.0.5. Composition in the homotopy category of topological spaces (Definition 2.0.4) is well-defined. If $f_1, f_2 : X \rightarrow Y$ are homotopic and $g_1, g_2 : Y \rightarrow Z$ are homotopic, then the compositions $g_1 \circ f_1$ and $g_2 \circ f_2$ are homotopic as maps from X to Z . That is, $[g] \circ [f] = [g \circ f]$ is independent of the choice of representatives.

Theorem 2.0.6. There exists a canonical functor $Q : \mathbf{Top} \rightarrow \mathbf{hTop}$ which is the identity on objects and maps each continuous map f to its homotopy class $[f]$. This functor is full and essentially surjective.

Definition 2.0.7 (Path in a topological space). Let X be a topological space (Definition 1.1.1), let $x_0, x_1 \in X$, and let $[0, 1] \subset \mathbb{R}$ be the closed unit interval equipped with the subspace topology from $\mathbb{R}$.

1. A *path in X from x_0 to x_1* is a continuous map $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. More generally, a *path in X* is a path from some point $x_0 \in X$ to some point $x_1 \in X$ in this sense.
2. A *loop in X based at $x_0 \in X$* is a path $\gamma : [0, 1] \rightarrow X$ in X from x_0 to x_0 , that is, a continuous map γ such that $\gamma(0) = \gamma(1) = x_0$. When a basepoint x_0 is fixed and understood from the context, a *loop in X* means a loop in X based at this chosen basepoint.

Definition 2.0.8 (Homotopy groups). For any pointed topological space (Definition 1.3.3) (X, x_0) and integer $n \geq 0$, the *n -th homotopy group of X at x_0* , denoted $\pi_n(X, x_0)$, is defined as the set of all homotopy classes (rel. ∂I^n) (Definition 2.0.3) of based maps

$$f : (I^n, \partial I^n) \rightarrow (X, x_0),$$

where $I^n = [0, 1]^n$. For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of based maps, and for $n \geq 2$, it is abelian.

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops (Definition 2.0.7) $\gamma : [0, 1] \rightarrow X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

Definition 2.0.9 (Weak homotopy equivalence). Let $f : X \rightarrow Y$ be a continuous map (Definition 1.3.1) between topological spaces. The map f is a *weak homotopy equivalence* if, for every choice of basepoint $x_0 \in X$ with $y_0 = f(x_0)$, the induced maps on homotopy groups (Definition 2.0.8)

$$f_* : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

are isomorphisms (Definition C.0.2) for all integers $n \geq 0$.

3. SINGULAR HOMOLOGY AND COHOMOLOGY

Definition 3.0.1. Let V be a real vector space of finite dimension. A *k -simplex in topology* (or a *geometric k -simplex*) is the convex hull of $k + 1$ affinely independent points $v_0, v_1, \dots, v_k \in V$, and is denoted by

$$[v_0, v_1, \dots, v_k] := \left\{ \sum_{i=0}^k t_i v_i \mid t_i \geq 0, \sum_{i=0}^k t_i = 1 \right\}.$$

It is also standard to talk of the *standard topological n -simplex* — the topological space, often denoted by notations such as Δ^n or $|\Delta^n|$ defined as the subset of Euclidean space $\mathbb{R}^{n+1}$ given by

$$|\Delta^n| = \left\{ (t_0, t_1, \dots, t_n) \in \mathbb{R}^{n+1} : \sum_{i=0}^n t_i = 1, \text{ and } t_i \geq 0 \text{ for all } i \right\}$$

equipped with the induced topology from the usual Euclidean topology on $\mathbb{R}^{n+1}$.

(♠ TODO: comment on how $|\Delta^n|$ makes sense via a geometric realization)

Definition 3.0.2 (Singular simplices). Let X be a topological space (Definition 1.1.1). For each integer $n \geq 0$, A *singular n -simplex in X* is a continuous map (Definition 1.3.1)

$$\sigma : \Delta^n \rightarrow X.$$

where Δ^n is the standard topological n -simplex (Definition 3.0.1). The set of all singular n -simplices in X is denoted by $S_n(X)$.

Definition 3.0.3 (Singular chain group with coefficients). Let X be a topological space (Definition 1.1.1), let $A \subseteq X$ be a subspace (Definition 1.1.8), and let $S_n(X)$ be the set of singular n -simplices in X (Definition 3.0.2).

1. Let R be a ring.

(a) The *singular n -chain group of X with coefficients in R* is the free R -module $C_n(X; R)$ whose elements are finite formal linear combinations

$$\sum_i r_i \sigma_i, \quad \text{with } \sigma_i \in S_n(X), r_i \in R.$$

Elements of $C_n(X; R)$ are called *singular n -chains in X with coefficients in R* .

(b) If $A \subseteq X$ is a subspace, the quotient groups

$$C_n(X, A; R) = C_n(X; R) / C_n(A; R)$$

may be referred to as the *relative singular n -chain groups of X in A with coefficients in R* and elements of this group may be referred to as *relative singular n -chains in X relative to A with coefficients in R* .

In either case, when $R = \mathbb{Z}$, the ring R may be suppressed from notation, so we may write $C_n(X)$ and $C_n(X, A)$ for $C_n(X, \mathbb{Z})$ and $C_n(X, A, \mathbb{Z})$ respectively.

2. Let G be an abelian group (Definition C.0.1).

(a) More generally, the *singular n -chain group of X with coefficients in G* is the abelian group

$$C_n(X; G) = C_n(X; \mathbb{Z}) \otimes_{\mathbb{Z}} G.$$

(??) In particular, an element of $C_n(X; G)$ is a finite formal sum

$$\sum_i g_i \sigma_i, \quad g_i \in G, \sigma_i \in S_n(X).$$

Elements of $C_n(X; G)$ are called *singular n -chains in X with coefficients in G* .

(b) Similarly, the *relative singular n -chain group of X in A with coefficients in G* is the quotient group

$$C_n(X, A; G) = C_n(X; G) / C_n(A; G)$$

Elements of $C_n(X, A; G)$ are called *singular n -chains in X relative to A with coefficients in G* .

When $G = R$ is a ring, note that the notions of $C_n(X; R)$ where R is taken as a ring and where R is taken as an abelian group are equivalent; similarly for $C_n(X, A; R)$.

We note that, in general $C_n(X; G) \cong C_n(X, \emptyset; G)$.

Definition 3.0.4 (Singular chain complex with coefficients). Let X be a topological space (Definition 1.1.1), let $A \subseteq X$ be a subspace (Definition 1.1.8), and let G be an abelian group.

For each $n \geq 1$, define the *boundary operator*

$$\partial_n : C_n(X; G) \rightarrow C_{n-1}(X; G)$$

(Definition 3.0.3) by

$$\partial_n(\sigma) = \sum_{i=0}^n (-1)^i \sigma \circ \delta_i,$$

where $\delta_i : \Delta^{n-1} \rightarrow \Delta^n$ is the i -th face inclusion, and extend ∂_n to $C_n(X; G)$ linearly. For $n \leq 0$, we simply let $\partial_n = 0$

One can verify that $(C_n(X; G), \partial_n)$ forms a chain complex (Definition E.0.1) in the abelian category of abelian groups. This chain complex is called the *(absolute) singular chain complex of X with coefficients in G* and is often denoted by $C_\bullet(X; G)$.

Then the boundary maps above induce maps

$$C_n(X, A; G) \rightarrow C_{n-1}(X, A; G)$$

on the relative chain groups $C_n(X, A; G)$, yielding a chain complex of abelian groups whose boundary maps, often also denoted by ∂_n , are induced by the boundary operators ∂_n above; this chain complex may be called the *relative singular chain complex of the pair (X, A) with coefficients in G* and is often denoted by $C_\bullet(X, A; G)$.

In the case that $G = R$ is a (not necessarily commutative) ring, then the boundary maps of $C_\bullet(X; R)$ and $C_\bullet(X, A; R)$ are left R -module homomorphisms:

$$\partial \left(\sum_i r_i \sigma_i \right) = \sum_i r_i \partial(\sigma_i)$$

Definition 3.0.5 (Singular homology with coefficients). Let X be a topological space (Definition 1.1.1), let $A \subseteq X$ be a subspace (Definition 1.1.8), and let G be an abelian group.

The *(absolute) n -th singular homology group of X with coefficients in G* is the homology group (Definition E.0.2)

$$H_n(X; G) = H_n(C_\bullet(X; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_\bullet(X; G)$ is the singular chain complex of X with coefficients in G and $\partial_n : C_n(X; G) \rightarrow C_{n-1}(X; G)$ are the boundary maps.

Given a subspace (Definition 1.1.8) $A \subseteq X$, the *n -th relative singular homology group of (X, A) with coefficients in G* is defined as

$$H_n(X, A; G) = H_n(C_\bullet(X, A; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_\bullet(X, A; G)$ is the relative singular chain complex of (X, A) with coefficients in G and $\partial_n : C_n(X, A; G) \rightarrow C_{n-1}(X, A; G)$ are the boundary maps.

We may denote $H_n(X; \mathbb{Z})$ and $H_n(X, A; \mathbb{Z})$ by $H_n(X)$ and $H_n(X, A)$ respectively.

In the case that $G = R$ is a ring, $H_n(X; R)$ and $H_n(X, A; R)$ have the structure of left R -modules (Definition D.0.2) since the corresponding chain complexes are chain complexes of left R -modules..

Definition 3.0.6 (Singular cochain complex of a topological space). Let X be a topological space (Definition 1.1.1) and R a commutative ring (Definition D.0.1) with 1. The *singular n -cochain group of X with coefficients in R* is the R -module

$$C^n(X; R) = \text{Hom}_R(C_n(X; R), R).$$

(Definition D.0.3) (Definition 3.0.3) We define *coboundary operators*

$$\delta^n : C^n(X; R) \rightarrow C^{n+1}(X; R), \quad \delta^n(\varphi) = \varphi \circ \partial_{n+1}.$$

The pair $(C^n(X; R), \delta^n)_n$ form a cochain complex (Definition E.0.1) called the *singular cochain complex of X with coefficients in R* . Given a subspace $A \subseteq X$, the *relative singular n -cochain group of (X, A) with coefficients in R* is the R -module

$$C^n(X, A; R) = \text{Hom}_R(C_n(X, A; R), R).$$

The coboundary operators δ^n induce maps $C^n(X, A; R) \rightarrow C^{n+1}(X, A; R)$ for which the $C^n(X, A; R)$ form a cochain complex, called the *relative singular cochain complex of (X, A) with coefficients in R*

Definition 3.0.7. Let X be a topological space (Definition 1.1.1) and R a commutative ring (Definition D.0.1) with 1. The *n -th singular cohomology group of X with coefficients in R* is

$$H^n(X; R) = \ker(\delta^n) / \text{im}(\delta^{n-1}).$$

For a subspace $A \subseteq X$ the *relative cohomology groups of (X, A)* are

$$H^n(X, A; R) = H^n(C^*(X, A; R)).$$

In other words, these singular cohomology groups are the cohomology R -modules (Definition E.0.2) of the singular cochain complex $C^*(X; R)$ and the relative singular cochain complex $C^*(X, A; R)$ (Definition 3.0.6) respectively.

APPENDIX A. SET THEORY

Definition A.0.1. Let A be a set.

- The set A is said to be *countably infinite* (or simply *countable*) if there exists a bijection $f : \mathbb{N} \rightarrow A$.
- The set A is said to be *finite* if there exists some $n \in \mathbb{N}$ and a bijection $g : \{1, 2, \dots, n\} \rightarrow A$.
- The set A is said to be *at most countable* if it is either finite or countably infinite.
- The set A is said to be *uncountable* if it is not at most countable.

Definition A.0.2. Let X and Y be sets and let $f : X \rightarrow Y$ be a function.

- The function f is said to be *injective* (or *one-to-one*) if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.

- The function f is said to be **surjective** (or **onto**) if for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- The map f is **bijective** if it is both injective and surjective. In this case, there exists a unique **inverse map** $f^{-1} : Y \rightarrow X$ such that for all $x \in X$ and $y \in Y$,

$$f^{-1}(f(x)) = x \text{ and } f(f^{-1}(y)) = y.$$

APPENDIX B. CATEGORY THEORY

Definition B.0.1 (Category). A **category** $\mathcal{C}$ consists of the following data:

- A class of **objects** denoted $\text{Ob}(\mathcal{C})$.
- For each pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, a class

$$\text{Hom}_{\mathcal{C}}(X, Y)$$

of **morphisms** (also called **arrows** or **homs**). If the category $\mathcal{C}$ is clear, then this **hom-class** is also denoted by $\text{Hom}(X, Y)$. It may also be denoted by $\text{hom}_{\mathcal{C}}(X, Y)$ or $\text{hom}(X, Y)$, especially to distinguish from other types of hom's (e.g. internal hom's)

- For each triple of objects X, Y, Z , a composition law

$$\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z),$$

denoted $(g, f) \mapsto g \circ f$.

- For each object X , an **identity morphism**

$$\text{id}_X \in \text{Hom}_{\mathcal{C}}(X, X).$$

These data satisfy the following axioms:

- (Associativity) For all morphisms $f \in \text{Hom}_{\mathcal{C}}(X, Y)$, $g \in \text{Hom}_{\mathcal{C}}(Y, Z)$, and $h \in \text{Hom}_{\mathcal{C}}(Z, W)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- (Identity) For all $f \in \text{Hom}_{\mathcal{C}}(X, Y)$,

$$\text{id}_Y \circ f = f = f \circ \text{id}_X.$$

One often writes $X \in \mathcal{C}$ synonymously with $X \in \text{Ob}(\mathcal{C})$, i.e. to denote that X is an object of $\mathcal{C}$.

We may call a category as above an **ordinary category** to distinguish this notion from the notions of *categories enriched in monoidal categories* or higher/ n -categories. (♠ **TODO: define n -categories**)

A category as defined above may be called a **large category** or a **class category** to emphasize that the hom-classes may be proper classes rather than sets (note, however, that the possibility that hom-classes are sets is not excluded for large categories). Accordingly, a **category** may often refer to a locally small category (Definition B.0.2), which is a category whose hom-classes are all sets.

Definition B.0.2 (Locally small category). A (large) category (Definition B.0.1) $\mathcal{C}$ is called a *locally small category* if for every pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, the collection $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a (small) *set* (as opposed to a proper class). In other words, each hom-class is a set and may even be called a *hom-set*.

In some contexts, a locally small category may simply be called a *category*, especially when genuinely large categories are not considered.

A category $\mathcal{C}$ is called a *small category* if it is a locally small category and the class $\text{Ob}(\mathcal{C})$ of objects is a set.

Remark B.0.3. Many “concrete” categories considered in “classical mathematics” or outside of more “abstract” category theory tend to be locally small. For example, the categories of sets, groups, R -modules, vector spaces, topological spaces, schemes, manifolds, sheaves on “small enough” sites are all locally small.

APPENDIX C. GROUP THEORY

Definition C.0.1 (Groups). A *group* is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a binary operation, subject to the following conditions:

1. (Associativity) For all $g, h, k \in G$ one has

$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$

2. (Identity element) There exists an element $e \in G$ such that for all $g \in G$,

$$e \cdot g = g \cdot e = g.$$

3. (Inverse element) For all $g \in G$ there exists an element $g^{-1} \in G$ such that

$$g \cdot g^{-1} = g^{-1} \cdot g = e.$$

The element e is called the *identity element of G* , and g^{-1} is called the *inverse of g* .

Equivalently, a group is a monoid with inverse elements.

A group $(G, \cdot)$ is often simply written as G , when the notation for the binary operation $\cdot$ is clear.

An *abelian group* or synonymously, a *commutative group*, is a group $(G, \cdot)$ whose binary operation $\cdot$ is *abelian* or *commutative*, i.e. satisfies

$$g \cdot h = h \cdot g$$

for all $g, h \in G$.

An abelian group is equivalent to a $\mathbb{Z}$ -module.

Definition C.0.2 (Group homomorphism). Let $(G, \cdot)$ and $(H, *)$ be groups (Definition C.0.1). A map $f : G \rightarrow H$ is called a **group homomorphism** if for all $g_1, g_2 \in G$ one has

$$f(g_1 \cdot g_2) = f(g_1) * f(g_2).$$

The collection of all groups with the group homomorphisms forms a locally small (Definition B.0.2) category (Definition B.0.1), called the **category of groups**.

If f is bijective (Definition A.0.2), then f is called a **group isomorphism**.

APPENDIX D. RING THEORY

Definition D.0.1. A **commutative (unital) ring** is a ring $(R, +, \cdot)$ such that $\cdot$ is a commutative operation, i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. “commutative” algebraists or number theorists), a **ring** refers to a commutative ring as above.

Definition D.0.2. Let R be a not-necessarily commutative ring.

1. A **left R -module** is an abelian group $(M, +)$ together with an operation $R \times M \rightarrow M$, denoted $(r, m) \mapsto rm$, such that for all $r, s \in R$ and $m, n \in M$:
 - $r(m + n) = rm + rn$,
 - $(r + s)m = rm + sm$,
 - $(rs)m = r(sm)$,
 - $1_R m = m$ where 1_R is the multiplicative identity of R .
2. A **right R -module** is defined similarly as an abelian group $(M, +)$ with an operation $M \times R \rightarrow M$, denoted $(m, r) \mapsto mr$, such that for all $r, s \in R$ and $m, n \in M$:
 - $(m + n)r = mr + nr$,
 - $m(r + s) = mr + ms$,
 - $m(rs) = (mr)s$,
 - $m1_R = m$.
3. Let R and S be (not necessarily commutative) rings.

An **R - S -bimodule** (or an **R - S -module** or an (R, S) -module, etc.) is an abelian group (Definition C.0.1) $(M, +)$ equipped with

(a) a left action of R :

$$R \times M \rightarrow M, \quad (r, m) \mapsto r \cdot m,$$

making M a left R -module (Definition D.0.2),

(b) a right action of S :

$$M \times S \rightarrow M, \quad (m, s) \mapsto m \cdot s,$$

making M a right S -module,

such that the left and right actions commute; that is, for all $r \in R$, $s \in S$, and $m \in M$,

$$r \cdot (m \cdot s) = (r \cdot m) \cdot s.$$

4. A **two-sided R -module** (or **R -bimodule**) is an R - R -bimodule.

If R is a commutative ring (Definition D.0.1), then a left/right R -module can automatically be regarded as a two-sided R -module. As such, we simply talk about **R -modules** in this case.

Any abelian group is equivalent to a two-sided $\mathbb{Z}$ -module. Moreover, any left R -module is equivalent to an $R - \mathbb{Z}$ -bimodule (Definition D.0.2) and any right R -module is equivalent to an $\mathbb{Z} - R$ -bimodule (Definition D.0.2). Given a left/right/two-sided R -module, its ***natural bimodule structure*** will refer to its structure as a $R - \mathbb{Z} / \mathbb{Z} - R / R - R$ bimodule. In this way, many definitions associated with the notions of left/right/two-sided R -modules can be defined as special cases for definitions for R - S -bimodules.

Definition D.0.3 (Hom of left/right/bi-modules). Let R, S, T be (not necessarily commutative) rings.

1. Let M and N be left R -modules (Definition D.0.2). The ***homomorphism group of left R -modules from M to N*** is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_R(M, N) = \text{Hom}_{R-\mathbb{Z}}(M, N) := \{f : M \rightarrow N \mid f \text{ is a left } R\text{-module homomorphism}\}.$$

2. Let M and N be right R -modules (Definition D.0.2). The ***homomorphism group of right R -modules from M to N*** is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_R(M, N) = \text{Hom}_{\mathbb{Z}-R}(M, N) := \{f : M \rightarrow N \mid f \text{ is a right } R\text{-module homomorphism}\}.$$

3. Let S be a (not necessarily commutative ring) and let M and N be $R - S$ -bimodules (Definition D.0.2). The ***homomorphism group of R - S -bimodules from M to N*** is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_{R-S}(M, N) := \{f : M \rightarrow N \mid f \text{ is a } R - S\text{-bimodule homomorphism}\}$$

In each case, $\text{Hom}(M, N)$ has a natural structure of an ***abelian group*** given by ***pointwise addition***: for $f, g \in \text{Hom}(M, N)$,

$$(f + g)(m) := f(m) + g(m),$$

and the zero morphism **0** given by $0(m) := 0_N$ acts as the identity element. The additive inverse **$-f$** is defined by $(-f)(m) := -f(m)$. Moreover, depending on bi-module structures that M and N may be carrying, $\text{Hom}(M, N)$ may itself carry additional module structures:

- In case that M is a $R - S$ -bimodule and N is a $R - T$ -bimodule, $\text{Hom}_R(M, N)$, the group of left R -module homomorphisms, is an $S - T$ -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(m \cdot s) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

- Dually, in case that M is a $S - R$ -bimodule and N is a $T - R$ -bimodule, $\text{Hom}_R(M, N)$, the group of right R -module homomorphisms, is an $S - T$ -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(s \cdot m) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

Some cases of interest may be when R, S , or T is in fact $\mathbb{Z}$ — these allow us to see module structures on $\text{Hom}(M, N)$ even when M and N are one-sided modules.

(♠ TODO: state this as a theorem) We furthermore note that $\text{Hom}_R(-, -)$ yields biadditive functors

$$\begin{aligned}\text{Hom}_R(-, -) &: {}_R\mathbf{Mod}_S^{\text{op}} \times {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T \\ \text{Hom}_R(-, -) &: {}_S\mathbf{Mod}_R^{\text{op}} \times {}_T\mathbf{Mod}_R \rightarrow {}_S\mathbf{Mod}_T.\end{aligned}$$

APPENDIX E. HOMOLOGICAL ALGEBRA

Definition E.0.1 (Chain complex in a preadditive category). Let $\mathcal{A}$ be a preadditive category and let I be a totally ordered set (typically $\mathbb{Z}$, but $I \subseteq \mathbb{Z}$ is also allowed).

1. A **chain complex** $(K_\bullet, d_\bullet)$ in $\mathcal{A}$ indexed by I is the homological convention for sequences with decreasing degrees. It consists of:

- Objects $\{K_i\}_{i \in I}$ in $\mathcal{A}$, called the **terms in degree i** ,
- Morphisms $d_i : K_i \rightarrow K_{i-1}$ in $\mathcal{A}$, called the **boundary maps** or **differentials in degree i** ,

such that for every $i \in I$, $d_{i-1} \circ d_i = 0$. That is,

$$K_\bullet : \dots \xrightarrow{d_{i+1}} K_i \xrightarrow{d_i} K_{i-1} \xrightarrow{d_{i-1}} K_{i-2} \rightarrow \dots$$

with $d_{i-1}d_i = 0$ for all i . We typically use the notation $K_\bullet = (K_i, d_i)_{i \in I}$.

2. Dually, a **cochain complex** $(K^\bullet, d^\bullet)$ in $\mathcal{A}$ follows the cohomological convention with increasing degrees. It consists of objects $\{K^i\}_{i \in I}$ and **coboundary maps** $d^i : K^i \rightarrow K^{i+1}$ such that $d^{i+1} \circ d^i = 0$:

$$K^\bullet : \dots \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \xrightarrow{d^{i+1}} K^{i+2} \rightarrow \dots$$

We typically use the notation $K^\bullet = (K^i, d^i)_{i \in I}$.

3. Let $K_\bullet = (K_i, d_i^K)$ and $L_\bullet = (L_i, d_i^L)$ be chain complexes (Definition E.0.1) in $\mathcal{A}$ indexed by the same set I . A **morphism of chain complexes** (or **chain map**)

$$f_\bullet : K_\bullet \rightarrow L_\bullet$$

consists of morphisms $f_i : K_i \rightarrow L_i$ for all $i \in I$, such that for every $i \in I$, the following diagram commutes:

$$\begin{array}{ccc} K_i & \xrightarrow{d_i^K} & K_{i-1} \\ \downarrow f_i & & \downarrow f_{i-1} \\ L_i & \xrightarrow{d_i^L} & L_{i-1} \end{array}$$

i.e., $d_i^L \circ f_i = f_{i-1} \circ d_i^K$.

A **morphism of cochain complexes** $f^\bullet : K^\bullet \rightarrow L^\bullet$ is defined similarly, satisfying the commutativity condition $d_L^i \circ f^i = f^{i+1} \circ d_K^i$.

The collection of these objects and morphisms forms a category. Notation for these categories is as follows:

- $\mathbf{Ch}(\mathcal{A})$ or $\mathbf{Ch}(\mathcal{A})$ is often used as a general term.
- To be explicit about the indexing convention, one uses $\mathbf{Ch}_\bullet(\mathcal{A})$ for chain complexes and $\mathbf{Ch}^\bullet(\mathcal{A})$ (or sometimes $\mathbf{CoCh}(\mathcal{A})$) for cochain complexes.

- The set of chain maps between two complexes is denoted by $\mathrm{Hom}_{\mathrm{Ch}(\mathcal{A})}(K_\bullet, L_\bullet)$; it is an abelian group under pointwise addition $(f + g)_i = f_i + g_i$.

Notations used to write chain complexes are *homological conventions* whereas notations used to write cochain complexes are *cohomological conventions*. In homological conventions, indices are written as subscripts and decrease when “following the differential maps”. In cohomological conventions, indices are written as superscripts and increase when “following the differential maps”. The theories of chain complexes and of cochain complexes are not genuinely different, however, as the conventions are essentially notational — starting from a chain complex $(K_\bullet, d_\bullet)$, one can obtain a cochain complex $(K^\bullet, d^\bullet)$ simply by defining $K^i = K_{-i}$ and d^i as d_{-i} and vice versa. As such, cochain complexes may be simply called *chain complexes* when careful distinctions between the conventions are unnecessary. Homological and cohomological notational conventions persist throughout homological algebra, see e.g. spectral sequences, double complexes, etc, but again the distinction between such conventions are merely notational.

Definition E.0.2 (Chain complexes and their (co)homology objects). Let $\mathcal{A}$ be an abelian category.

- For a cochain complex $K^\bullet$, its *cohomology object in degree i* is defined as the quotient of the object of i -cocycles by the object of i -coboundaries:

$$H^i(K^\bullet) := Z^i(K)/B^i(K) = \ker(d^i)/\mathrm{im}(d^{i-1}).$$

In fact, each H^i is a functor $H^i : \mathbf{Ch}^\bullet(\mathcal{A}) \rightarrow \mathcal{A}$ (Definition E.0.1); given a morphism $f^\bullet : K^\bullet \rightarrow L^\bullet$ of cochain complexes, it is thus customary to write $H^i(f^\bullet) : H^i(K^\bullet) \rightarrow H^i(L^\bullet)$ for the induced morphism on the i th cohomology objects.

- For a chain complex $K_\bullet$, its *homology object in degree i* is defined as the quotient of the object of i -cycles by the object of i -boundaries:

$$H_i(K_\bullet) := Z_i(K)/B_i(K) = \ker(d_i)/\mathrm{im}(d_{i+1}).$$

In fact, each H_i is a functor $H_i : \mathbf{Ch}_\bullet(\mathcal{A}) \rightarrow \mathcal{A}$ (Definition E.0.1); given a morphism $f_\bullet : K_\bullet \rightarrow L_\bullet$ of chain complexes, it is thus customary to write $H_i(f_\bullet) : H_i(K_\bullet) \rightarrow H_i(L_\bullet)$ for the induced morphism on the i th homology objects.

See ??

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